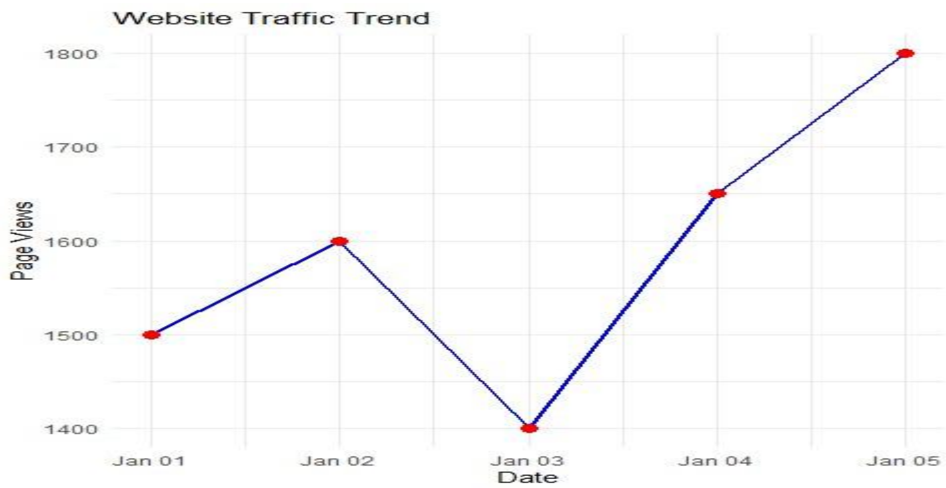
SET 21 - Website Traffic Analysis

21 A. Line Chart

Program:

```
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03",
                   "2023-01-04",
                   "2023-01-05")),
  PageViews =
c(1500,1600,1400,1650,1800),
CTR = c(2.3,2.7,2.0,2.4,2.6))
ggplot(traffic_data,     aes(x=Date,
y=PageViews,         group=1))+
geom_line(color="blue",linewidth=1
)+
geom_point(color="red",size=3)+
labs(title="Website Traffic Trend",
x="Date",       y="Page Views")+
theme_minimal()
```

Output:



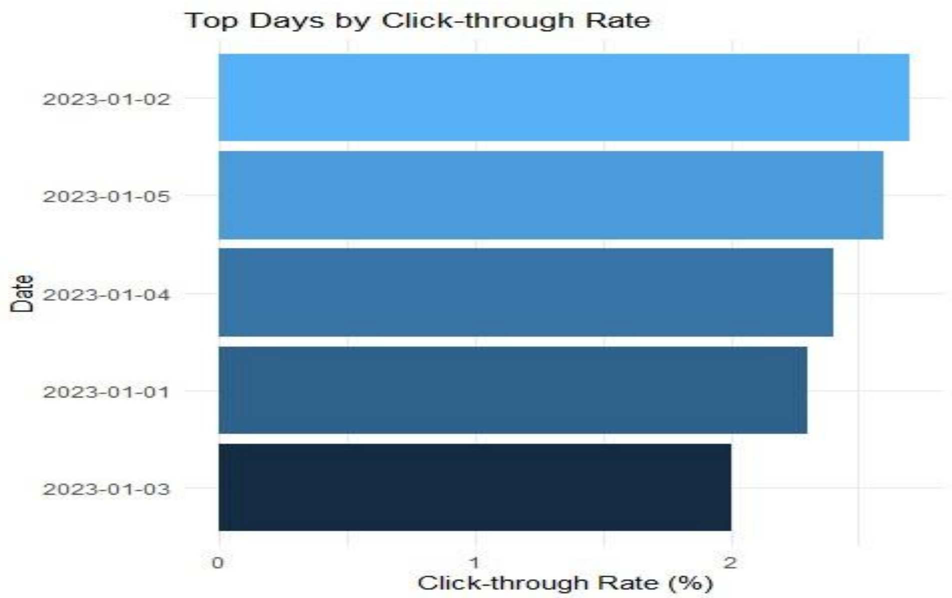
21 B. Bar Chart

Program:

```
top_ctr <- traffic_data[order(-
traffic_data$CTR),] ggplot(top_ctr,
aes(x=reorder(as.character(Date),CTR),
```

```
y=CTR,          fill=CTR))+
geom_bar(stat="identity")+
coord_flip()+   labs(title="Top Days by
Click-through Rate",        x="Date",
y="Click-through Rate (%)")+
theme_minimal()+
theme(legend.position="none")
```

Output:



21 C. Stacked Area Chart

Program:

```
interaction_data <- data.frame(
  Date=as.Date(c("2023-01-01",
                 "2023-01-02",
                 "2023-01-03",
                 "2023-01-04",
                 "2023-01-05")),
  Likes=c(120,150,110,170,180),
  Shares=c(60,75,55,80,90),
Comments=c(30,40,25,45,50))
id.vars="Date",
variable.name="Interaction",
value.name="Count")
ggplot(interaction_long,
aes(x=Date,y=Count,
```

```
        fill=Interaction,
group=Interaction))+
geom_area(alpha=0.8)+
labs(title="Distribution of User
Interactions",      x="Date",
y="Interaction Count")+
theme_minimal()
```
Output: